Polar Form, De Moivre's Theorem & Roots of Unity

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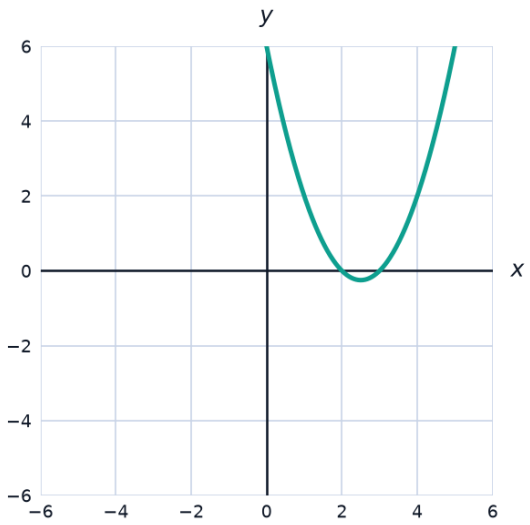
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Use De Moivre's theorem to prove that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Hence solve $4x^3 - 3x = 0$ on $x \in [-1, 1]$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = x^2 - 5x + 6$. The graph of this function is a parabola. Determine the x-coordinates of the points where the graph intersects the x-axis.



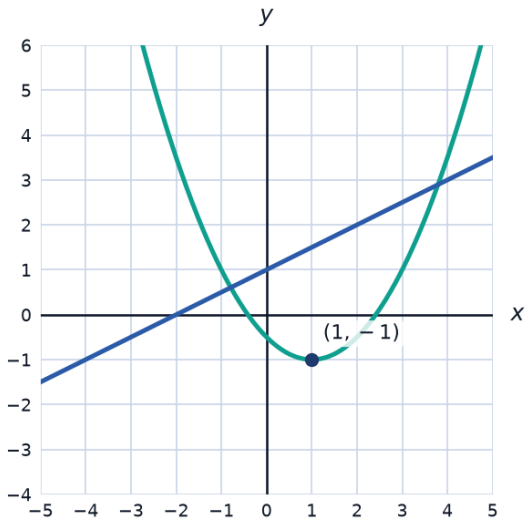
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = ax^2 + bx + c$. The graph of $f(x)$ is a parabola that passes through the points $(-1, 12)$, $(1, 2)$, and $(2, 9)$. Find the value of the parameter b.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that L_1 and L_2 are skew, and find the shortest perpendicular distance between them.



5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

(IB AI HL) A market transition matrix $T = (0.7, 0.2; 0.3, 0.8)$ represents weekly customer movement between Brand A and Brand B. Find: (a) the probability that a customer initially with Brand A is with Brand B after 2 weeks; (b) the long-term steady-state market distribution.

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Determine the equation of the oblique asymptote to the graph of $f(x) = \frac{2x^2 - 3x + 4}{x - 1}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use trigonometric substitution to evaluate the indefinite integral $\int \frac{1}{(4 + x^2)^{3/2}}\,dx$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

(IB AI HL) A communications network with 6 nodes $\{A, B, C, D, E, F\}$ has weighted edges: AB=4, AC=2, BC=1, BD=5, CD=8, CE=10, DE=2, DF=6, EF=3. Use Kruskal's algorithm to determine the minimum spanning tree, stating the order of edges added, and find the total tree weight.

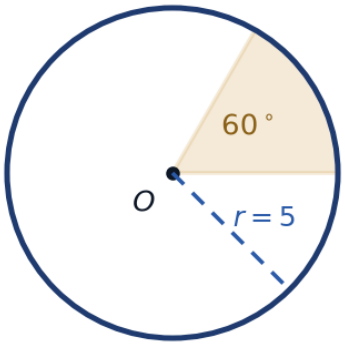
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = \frac{\sin(2x)}{\cos(x)}$. For the domain $x \in [0, 2\pi]$ such that $\cos(x) eq 0$, find the exact value of the definite integral $\int_0^{\pi/2} f(x)dx$.

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10 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Using the binomial series for $(1 - x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x) = \arcsin x$ up to and including the x^5 term. State its radius of convergence.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A survey was conducted on the number of hours students spent studying per week. The results are summarized in the histogram below. The total number of students surveyed is 120. Let x represent the number of students who studied for more than 3 hours but at most 4 hours. Find the value of x.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Decompose the rational expression $\frac{5x^2 - 3x + 2}{(x - 1)(x^2 + 1)}$ into partial fractions.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The derivative of the function $f(x) = 3x^2 - 5x + 2$ is given by $f'(x)$. Find the value of $f'(2)$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A random variable X follows a binomial distribution $B(n, p)$. The number of trials is $n = 10$ and the probability of success in a single trial is p. If the mean of X is 4, find the variance of X.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.03 Markov Chains (AI HL)

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Proof by Mathematical Induction

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.04 Partial Fractions

Use the substitution $x = 2\sin\theta$ to evaluate the exact definite integral $\int_0^1 \sqrt{4 - x^2}\, dx$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Find the exact range of $f(x) = \frac{x}{x^2 + 1}$ for $x \in \mathbb{R}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

A weighted network has edges: DE = 1, BC = 2, AB = 3, CD = 4, BD = 5, CE = 6, AC = 7, AD = 8. Use Kruskal's algorithm to determine the minimum spanning tree and compute its total weight.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $z = -2 - 2\sqrt{3}i$ in Euler's exponential form $re^{i\theta}$ with $\theta \in (-\pi, \pi]$.

a. $4e^{-i2\pi/3}$

b. $4e^{i4\pi/3}$

c. $2e^{-i2\pi/3}$

d. $4e^{-i\pi/3}$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert $z = 6e^{i3\pi/4}$ into exact Cartesian form $x + iy$.

a. $-3\sqrt{2} + 3\sqrt{2}i$

b. $3\sqrt{2} - 3\sqrt{2}i$

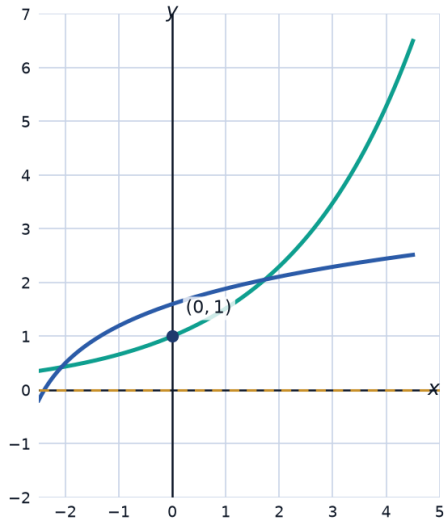
c. $-3 + 3\sqrt{3}i$

d. $-3\sqrt{2} - 3\sqrt{2}i$

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $z_1 = 3e^{i\pi/3}$ and $z_2 = 2e^{i\pi/6}$, evaluate the quotient $\frac{z_1}{z_2}$ in exponential form.



a. $\frac{3}{2}e^{i\pi/6}$

b. $\frac{3}{2}e^{i\pi/2}$

c. $6e^{i\pi/6}$

d. $1.5e^{-i\pi/6}$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Use De Moivre's Theorem to evaluate $(1 + i\sqrt{3})^6$.

a. 64

b. -64

c. $64i$

d. $-64i$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $\cos(3\theta)$ in terms of $\cos\theta$ using De Moivre's Theorem $(\cos\theta + i\sin\theta)^3$.

a. $4\cos^3\theta - 3\cos\theta$

b. $3\cos^3\theta - 4\cos\theta$

c. $\cos^3\theta - 3\cos\theta$

d. $4\cos^3\theta + 3\cos\theta$

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find all cube roots of $8i$ in polar form.

a. $2e^{i\pi/6}, 2e^{i5\pi/6}, 2e^{-i\pi/2}$

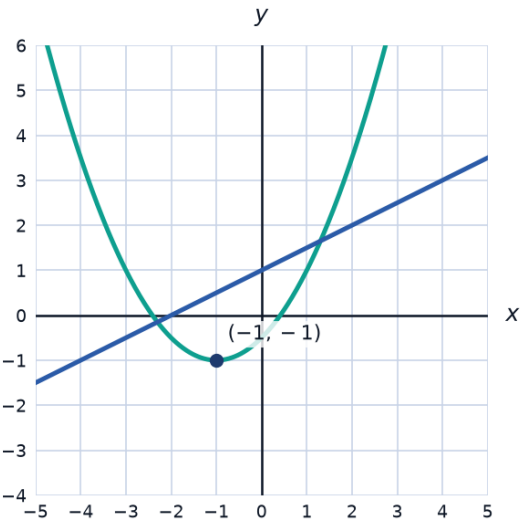
b. $2e^{i\pi/3}, 2e^{i\pi}, 2e^{i5\pi/3}$

c. $8e^{i\pi/6}, 8e^{i5\pi/6}, 8e^{-i\pi/2}$

d. $2, -1+i\sqrt{3}, -1-i\sqrt{3}$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How are the n distinct nth roots of any complex number $z = re^{i\theta}$ distributed on the Argand plane?



a. They lie on a circle of radius $r^{1/n}$ centered at the origin, evenly spaced by $\frac{2\pi}{n}$ radians

b. They lie along a straight line through the origin

c. They form an ellipse with eccentricity $1/n$

d. They cluster exclusively in Quadrant I

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the sum of all n distinct nth roots of any complex number z ($n \geq 2$)?

a. 0

b. 1

c. n

d. z

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

If $\omega = e^{i2\pi/3}$ is a primitive complex cube root of unity, what is the value of $1 + \omega + \omega^2$?

a. 0

b. 1

c. -1

d. 3

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Factor $z^4 + 1$ into two real quadratic factors.

a. $(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$

b. $(z^2 + 1)^2$

c. $(z^2 - 1)(z^2 + 1)$

d. $(z^2 - 2z + 1)(z^2 + 2z + 1)$

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The four 4th roots of unity ($z^4 = 1$) form the vertices of which geometric shape on the Argand plane?

a. A square inscribed in the unit circle with vertices at $\pm 1, \pm i$

b. An equilateral triangle

c. A regular hexagon

d. A rectangle of perimeter 4

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

For $x^2 - 6x + 5 = 0$, find the sum of the roots.

a. 6

b. 11

c. 5

d. 3

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33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Let the complex number z be represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Suppose z satisfies the equation $z^3 = -64i$. If z is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2} < \arg(z) < 2\pi$), find the real part of z.

a. 4

b. -4

c. $2\sqrt{3}$

d. $-2\sqrt{3}$

34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Let the complex number z_1 be represented in polar form as $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, where $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. Let another complex number, z_2, be represented as $z_2 = 2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$ and $\alpha = \frac{5\pi}{6}$, find the value of θ_2 in radians, given that $0 \le \theta_2 < 2\pi$.

a. $\frac{\pi}{2}$

b. $\frac{2\pi}{3}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{6}$

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \le \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$

b. $r = 1, \theta = \frac{4\pi}{3}$

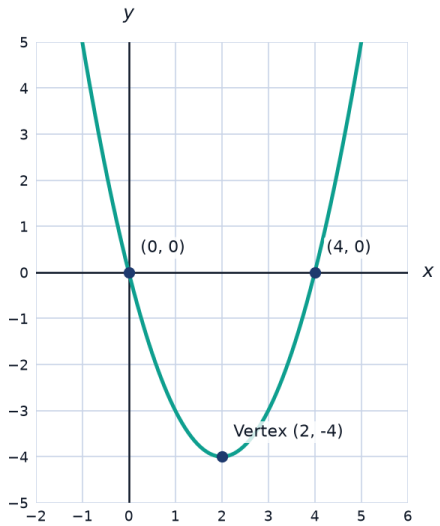
c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

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36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The graph of the function $y = f(x)$ is shown below. The function is of the form $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(2, -4)$. The parabola passes through the points $(0, 0)$ and $(4, 0)$. Determine the value of a.



a. -1

b. 1

c. -2

d. 2

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A complex number z is represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Given that $z^3 = 8i$, and z lies in the first quadrant such that its argument θ satisfies $0 < \theta < \frac{\pi}{2}$. Find the value of $r^2\cos\theta$.

a. 2

b. 4

c. $2\sqrt{2}$

d. 1

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.04 Partial Fractions

A complex number z is represented in polar form as $z = r(\cos(\theta) + i\sin(\theta))$. If a complex number has a modulus $r = 5$ and an argument $\theta = \frac{\pi}{3}$, what is its rectangular form $a + bi$?

a. $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$

b. $5 + i5$

c. $\frac{5\sqrt{3}}{2} + i\frac{5}{2}$

d. $5i$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Find the oblique asymptote of $f(x) = \frac{4x^2 + 3x - 14}{(x + 2)}$, in the form $y = mx + c$.

a. $y = 4x - 5$

b. $y = 6x - 4$

c. $y = 5x - 4$

d. $y = 3x - 4$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Area enclosed by $r = 2\cos\theta$ (polar):

a. π

b. 2π

c. 4π

d. $\pi/2$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Express $f(x) = \frac{x^2 - 3x + 5}{(x - 3)}$ in the form $mx + c + \frac{r}{(x - 3)}$, and state the value of r.

a. 8

b. 7

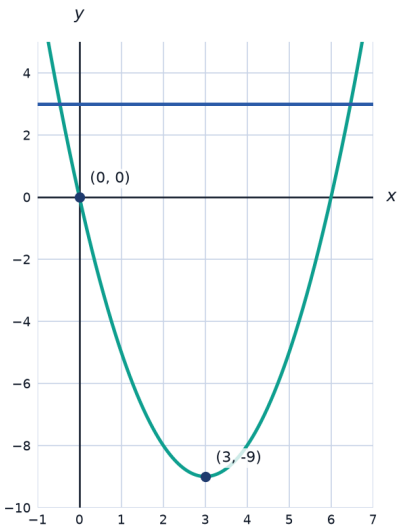
c. 2

d. 5

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

The graph of the function $f(x) = ax^2 - 6x + c$ is shown below. The graph has a vertex at the point (3, -9) and passes through the point (0, 0). Determine the values of a and c.



a. $a = 1, c = 0$

b. $a = 2, c = 0$

c. $a = 1, c = 3$

d. $a = 2, c = 3$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Consider the complex number $z = r(\cos\theta + i\sin\theta)$. If $z^3 = -8$ and $r > 0$, and the principal argument of z is $\frac{\pi}{3}$, find the value of r.

a. 2

b. 4

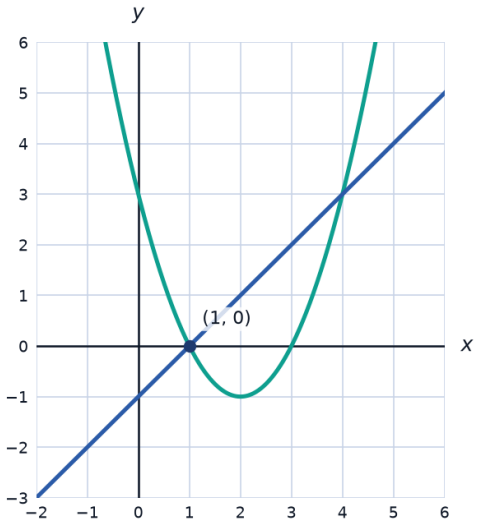
c. 8

d. 16

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

The diagram shows the graph of the function $y = f(x)$. The function is defined by $f(x) = x^2 + bx + c$. The graph passes through the point $(1, 0)$ and has a vertex at $(2, -1)$. Find the value of b.



a. -4

b. 4

c. -2

d. 2

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.02 The Poisson Distribution

The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx - 2$. The graph passes through the points $(1, -3)$ and $(2, 2)$. Determine the value of a.



a. 1

b. 2

c. 3

d. 4

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